Literature Review

(CSIT321 – Final Year Project)

CSIT-26-S2-05: A Mobile Application for Wise Workout

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1. Introduction & Scope

1.1 Purpose of This Review

In this review, we examine the current landscape of workout monitoring and fitness mobile applications, focusing on running and cycling. It brings together findings from research papers, existing commercial apps, and industry sources to answer the project brief's main questions and guide the design of our new fitness app.

1.2 Why Running and Cycling?

- Data richness:
  - Running and cycling generate the widest variety of measurable exercise data: heart rate, pace, power, cadence, GPS route, elevation, and stride metrics.
  - This allows our system to have the richest input space for analytics and personalisation algorithms.
- Sensor ecosystem maturity:
  - Both sports have well-established third-party sensor ecosystems built on BLE and ANT+ standards, covering heart rate monitors, foot pods, cadence sensors, power meters, and GPS devices.
  - This means data acquisition strategies are well-documented and reproducible.
- Social and competitive culture:
  - Running and cycling communities have strong participation cultures such as parkrun, Strava segments, club rides, and virtual races that map directly to the social and competition features required by the project specification.

1.3 Scope Definition

Dimension	In Scope	Out of Scope
Sports	Running, Cycling (outdoor & indoor)	Swimming, strength training, yoga, team sports
Devices	Smartphones, smartwatches, BLE/ANT+ sensors, smart trainers	Medical-grade devices, implantable monitors
Platform	iOS and Android mobile applications	Desktop applications, web-only tools
Users	Recreational and amateur athletes	Elite/professional athletes with specialised needs
Features	Data collection, analytics, coaching, social, gamification	Nutrition tracking, sleep monitoring (except where linked)

Table 1.1 — Scope boundaries for this literature review

2. Market Landscape & Competitor Analysis

2.1 Overview of the Fitness App Market

The global fitness app market was valued at approximately USD 12.12 billion in 2025 and is projected to reach USD 33.58 billion by 2033, driven by increasing health awareness, technological advances, and the continued adoption of digital fitness solutions (Grand View Research, 2026). Within this broader market, endurance sport applications, particularly those focused on running and cycling, represent commercially meaningful subsegments. For example, the global running app market was estimated at approximately USD 485 million in 2021 and is expected to continue expanding through 2030 (Verified Market Reports, 2025).

2.2 Key Applications in the Market

Among the reviewed applications, Strava shows the largest publicly disclosed user scale, with approximately 180 million registered users and nearly 50 million monthly active users in 2025. MapMyRun / MapMyFitness reports over 90 million users, while Garmin Connect reports over 25 million users. Public figures for the remaining platforms are limited.

App	Platform	Primary Data Sources	Profile Recommendations	Social / Competition	Business Model
Strava	iOS, Android, Web	GPS, BLE sensors, manual	Basic (training load, fitness score)	Very strong — segments, clubs, challenges	Freemium (Strava Summit)
Garmin Connect	iOS, Android	Garmin devices, BLE	Strong — VO2max, training status, FitSpark	Moderate — leaderboards, badges	Free (device-bundled)
Nike Run Club	iOS, Android	Phone GPS, Apple Watch	Coached audio runs, plans	Moderate — challenges, friend feed	Free
Wahoo SYSTM	iOS, Android, Web	Smart trainers, power meters	Very strong — 4DP profile, adaptive plans	Weak — no community features	Subscription ($14.99/mo)
Polar Flow	iOS, Android, Web	Polar devices, BLE HR	Strong — FitSpark daily suggestions	Weak	Free (device-bundled)
Zwift	iOS, Android, PC, Mac	Smart trainers, BLE/ANT+	Structured workouts, plans	Very strong — virtual racing	Subscription ($19.99/mo)
MapMyRun (Under Armour)	iOS, Android	GPS, phone sensors, BLE	Basic — shoe tracking, cadence tips	Moderate — challenges	Freemium
adidas Running (Runtastic)	iOS, Android	GPS, phone sensors	Basic — goal-based plans	Moderate	Freemium

Table 2.1 — Competitor feature comparison matrix

2.3 Feature Gap Analysis

Reviewing the competitor landscape reveals gaps that our SaaS application can address:

- Few apps combine personalised coaching with strong social competition features in a seamless user experience.
- Many profile-based recommendation features are placed behind premium tiers, which may reduce early user adoption and engagement.
- Cross-sport profile continuity across activities such as running and cycling is often inconsistent.
- Most fitness apps only send scheduled reminders, while smarter alerts for recovery and overtraining are much less common.

Summary:

- The market gap sits at the intersection of AI-personalised coaching, social gamification, and genuine cross-platform accessibility, as no single mainstream product clearly combines all three in one offering.

2.4 SWOT Analysis of Existing Market

STRENGTHS	WEAKNESSES
• Mature BLE/ANT+ sensor ecosystems • Strava's segment model proven at scale • Strong community of data-literate athletes • Cross-platform frameworks are now mature (Flutter)	• Battery drain from always-on GPS, etc. • Sensor accuracy varies by device tier • Social features risk privacy backlash • Premium conversion rates typically <10%
OPPORTUNITIES	**THREATS**
• AI coaching still underserved in the free tier • Post-COVID running participation surge • Integration with health platforms such as Apple Health and Google Fit • Gamified social challenges can improve retention and engagement • Unified multi-activity profiling remains underdeveloped in many apps • Smarter recovery and overtraining insights are still uncommon • Beginner-friendly personalised coaching remains underserved	• Apple/Google building native fitness features • Strava premium price sensitivity • Wearable platform lock-in (Garmin, Wahoo) • Data privacy regulation (GDPR, PDPA)

Table 2.2 — SWOT analysis of the fitness app market landscape

3. Data Collection: Sensors, Protocols & Devices

3.1 Exercise Data Streams in Running and Cycling

Running and cycling applications rely on multiple exercise data streams collected from smartphone sensors, wearable devices, and dedicated sports hardware. Common data types include GPS-based location and pace, heart rate, cadence, elevation, and speed, while more advanced metrics such as cycling power, stride length, ground contact time, and heart rate variability provide deeper performance insights. The precision of these measurements varies by device, with dedicated sensors such as chest straps, power meters, and barometric altimeters generally offering higher accuracy than phone- or watch-based estimates. Overall, these data streams form the basis for activity tracking, performance analysis, and personalised recommendations in modern endurance sport applications.

Data Type	Sensor / Source	Sports	Precision Level
GPS location & pace	Phone GPS chip or watch GPS	Running, Cycling	±3–5m (phone), ±1–2m (dedicated GPS)
Heart rate	Optical PPG (wrist) or chest strap ECG	Both	±2bpm optical; ±1bpm strap
Cadence (steps/rpm)	Accelerometer (running) / crank sensor (cycling)	Running, Cycling	±1 step or revolution
Cycling power	Smart trainer, pedal-based or crank-based power meter	Cycling	±1–2% (high-end meters)
Elevation	Barometric altimeter (watch) or GPS altitude	Both	±1m barometric; ±10m GPS
Speed (cycling)	Wheel speed sensor or GPS	Cycling	±0.5 km/h
Stride length & GCT	Foot pod or watch accelerometer	Running	±5cm stride; ±5ms GCT
HRV (Heart Rate Variability)	Chest strap or advanced watch PPG	Both	±5ms RMSSD

Table 3.1 — Exercise data streams by type, sensor, and sport

3.2 Communication Protocols

The data acquisition layer relies on standardised wireless protocols. The choice of protocol significantly affects battery life, range, multi-sensor compatibility, and implementation complexity.

Protocol	Range	Power Draw	Data Rate	Primary Use Case in Fitness
Bluetooth Low Energy (BLE 5.0)	~100m	Very low	2 Mbps	HR monitors, cadence sensors, shoe pods, most modern wearables
ANT+	~30m	Ultra-low	~60 kbps	Multi-device cycling (power + cadence + speed simultaneously)
Wi-Fi (802.11)	~50m	High	Up to 600 Mbps	Smart trainer sync, post-session data upload
NFC	<5cm	Passive (reader provides power)	~400 kbps	Shoe pods (select Adidas/Nike products), contactless pairing
GPS (passive receive)	Global (satellite)	Moderate (receiver only)	N/A — receive only	Route tracking, pace, elevation in outdoor sports

Table 3.2 — Wireless protocol comparison for fitness sensor applications

3.3 The GATT Standard and BLE Sensor Profiles

The Generic Attribute Profile (GATT) governs how BLE devices expose data to connected applications. Most running and cycling sensors use standardised GATT service profiles, which enables interoperability between devices from different manufacturers:

- Heart Rate Service (0x180D)
  - Standard profile for all BLE HR monitors
- Cycling Speed and Cadence Service (0x1816)
  - Wheel revolution and crank revolution data
- Cycling Power Service (0x1818)
  - Power in watts, torque, and pedal balance
- Running Speed and Cadence Service (0x1814)
  - Step rate and stride data from foot pods

Adherence to GATT profiles is what enables applications to connect to any compliant sensor without device-specific drivers, which is a critical implementation consideration for achieving broad device compatibility.

3.4 Data Pipeline Architecture

The following describes the data flow from sensor to user-facing analytics:

1. Sensor acquisition:
    - Raw measurements are broadcast by peripheral devices (HR monitor, power meter) over BLE/ANT+ at 1–4 Hz.
2. Central collection:
    - The smartphone or watch acts as the GATT central device, aggregating streams from multiple peripherals into a single session record.
3. Local processing:
    - Basic metrics (average HR, distance, pace) are computed on-device for real-time display with minimal latency.
4. Cloud upload:
    - On session completion, the full raw data stream is uploaded to a cloud backend via HTTPS, typically compressed as JSON or FIT file format.
5. Analytics engine:
    - Server-side processing computes derived metrics (training load, VO2max estimate, zone distribution) and updates the user's long-term fitness profile.
6. Delivery:
    - Processed results are returned to the app for display and stored in the user's profile database for longitudinal analysis.

4. Personalisation, Profiling & AI-Driven Recommendations

4.1 User Profiling Inputs

Effective personalisation requires a multi-layered user profile. Inputs can be categorised as static (captured at onboarding) and dynamic (updated continuously from activity data):

Profile Layer	Data Points	Update Frequency	Powers
Static biometric	Age, sex, weight, height, resting HR	On change	Calorie estimation, heart rate zone calculation, VO2max baseline
Goal state	Target event (5K, sportive), weekly distance goal, weight loss target	User-set	Plan generation, workout selection, progress benchmarks
Fitness baseline	Estimated VO2max, FTP (cyclists), VDOT (runners)	Post-activity update	Workout intensity targets, training load thresholds
Training history	Volume, intensity, consistency over past 6 weeks	Per session	Training load modelling, recovery estimation, overtraining detection
Behavioural	Preferred workout times, session frequency, skip patterns	Ongoing	Smart scheduling, reminder timing, plan adherence modelling

Table 4.1 — User profile layers and the analytics they enable

4.2 The ATL/CTL/TSB Training Load Model

These are among the dominant algorithms used for training analytics in endurance sport applications by Garmin, TrainingPeaks, Wahoo, and multiple academic performance prediction systems.

Metric	Also Known As	Formula	Interpretation
ATL (Acute Training Load)	Fatigue	Exponential weighted avg of daily TSS, τ=7 days	Short-term fatigue — how tired you are from recent training
CTL (Chronic Training Load)	Fitness	Exponential weighted avg of daily TSS, τ=42 days	Long-term fitness — your accumulated aerobic base
TSB (Training Stress Balance)	Form	CTL − ATL	Positive = fresh & ready; negative = fatigued; near zero = baseline
TSS (Training Stress Score)	Input metric	$(\text{Duration} \times \text{IF}^2) \times 100 / 3600$	A single number representing the training stress of one session

Table 4.2 — ATL/CTL/TSB model components and interpretation

Practically, our app can use these models when:

- TSB drops below −30(alert a user, high fatigue risk),
- ATL spikes 30%+ above CTL (recommend a rest day).
- TSB approaches +10 to +25 (peak performance zone; suggest a race window).

4.3 VO2max Estimation

VO2max (maximal oxygen uptake) is the gold standard measure of aerobic capacity. Consumer fitness apps estimate it non-invasively using one of three approaches:

- Submaximal HR-pace method (Firstbeat algorithm, used by Garmin and Polar):
  - Pace and HR from a normal run are regressed against known HR-VO2 relationships.
- Cooper test approximation:
  - A user's 12-minute running distance is mapped to a VO2max estimate.
- Resting HR method:
  - Tanaka formula using resting HR and age for a rough population-level estimate, used when no exercise data is available.

4.4 Machine Learning Approaches

Approach	Application	Example / Citation	Maturity
Collaborative filtering	Workout recommendations based on users with similar profiles and history	Spotify-style 'users like you completed X' logic	Commercial
Random forest regression	Predicting race finish time from training history features	Vickers & Vertosick (2016) — marathon time prediction	Academic
LSTM / Transformer	Sequential training load modelling over weeks/months to predict performance peaks	Burbidge et al. (2022) — deep learning endurance	Emerging
Anomaly detection (Isolation Forest)	Flagging unusual HR patterns that may indicate overtraining or health events	Used in smartwatch firmware (Apple Watch)	Commercial
Reinforcement learning	Adaptive plan generation that adjusts weekly targets based on user adherence	Research prototype stage	Research

Table 4.3 — Machine learning techniques applied to fitness personalisation

5. LLMs for Personalisation, Profiling & AI-Driven Recommendations

5.1 Background and Motivation

Traditional personalisation in fitness apps has relied on rule-based systems and classical machine learning, such as the ATL/CTL/TSB models, collaborative filtering for workout recommendations, and regression models for performance prediction. These approaches are effective but brittle as they require structured input, cannot handle ambiguous user intent, and produce outputs that feel mechanical rather than coached.

Using Large Language Models (LLMs), which are transformer-based models trained on vast corpora of text, introduces a fundamentally different capability. It enables systems to reason over unstructured context, generate natural language explanations, and synthesise heterogeneous information into coherent, personalised guidance.

Since 2022, the deployment of LLMs in consumer health and fitness contexts has accelerated significantly, and several patterns of application have emerged that are directly relevant to a running and cycling SaaS product.

5.2 How LLMs Differ from Classical Personalisation

Capability	Classical ML / Rule-based	LLM-based
Training load calculation	Strong: ATL / CTL / TSB is mathematically precise	Weak: Should not replace numerical models
Performance prediction	Strong: regression on structured features	Moderate: useful for reasoning about edge cases
Natural language coaching	Cannot do this	Strong: natural, context-aware explanation
Goal interpretation	Rigid: requires structured input fields	Strong: handles “I want to run a half marathon by October, but I’ve been injured”
Cross-domain synthesis	Weak: siloed per metric	Strong: synthesises HR, sleep, weather, schedule, history
Ambiguity handling	Poor: breaks on missing fields	Strong: handles incomplete profiles gracefully
Explainability of recommendation	None: outputs a number	Strong: can explain why a recommendation is being made

The practical implication is that the optimal architecture combines quantitative models that generate the underlying metrics (TSS, VO2max estimate, training load) with an LLM layer that interprets those numbers, contextualises them against the user's stated goals and history, and communicates them in natural language.

5.3 LLM Application Patterns in Running and Cycling

5.3.1 Conversational Coaching Interfaces

The most direct application is a chat-based coaching layer where the user can ask natural language questions about their training. Rather than navigating to a training load chart and interpreting numbers themselves, a user can ask: “Why does my coach plan say to take it easy today when I feel fine?”

The LLM can retrieve the user’s current TSB, recent session history, and any flagged HRV anomalies, and respond with something like: “Your training stress balance is quite negative after 3 hard days, so your body is accumulating fatigue faster than it is recovering. Even if you feel okay now, pushing hard today would likely degrade your performance for Saturday’s long run”.

This pattern of using retrieval-augmented generation (RAG) applied to a user's personal fitness data is the most commercially viable LLM application in fitness currently. Whoop introduced a version of this in 2023 with their AI coaching feature, and Apple Fitness+ has experimented with contextual coaching narration.

The key technical requirement is a well-structured retrieval layer: the LLM needs access to relevant user data (recent sessions, current metrics, training plan state) as context, not just general exercise science knowledge.

5.3.2 Goal Disambiguation and Plan Generation

The classical way of generating a plan requires users to fill structured forms with fields like goal event, target date, current weekly mileage, number of days available. Users frequently enter inaccurate information or do not understand what is being asked. LLMs enable a conversational onboarding flow where the user simply describes their situation in free text, and the model extracts the relevant parameters, asks clarifying questions where necessary, and negotiates a realistic plan.

For example, a user might type: *"I used to run a lot but haven't trained in about six months. I signed up for a 10K in March and I'm hoping to go sub-55 minutes."* An LLM can recognise the detraining implication, estimate a realistic starting fitness baseline, identify the timeline as approximately 14–18 weeks depending on when in the year this is asked, and generate a conservative return-to-running plan that manages injury risk — while explaining the reasoning to the user in plain language. This is qualitatively different from a dropdown-based form.

5.3.3 Contextual Recommendation Enrichment

LLMs can enrich recommendations that would otherwise be delivered as bare numbers or generic instructions. When a classical system determines that today's prescribed workout is a Zone 2 easy run of 45 minutes, an LLM layer can add: the physiological rationale for Zone 2 training, a reminder that the weather forecast shows 28°C heat requiring pace adjustment, a note that the user's last easy run was paced too hard based on HR data, and a specific pacing target in the user's preferred units. Each of these enrichments draws from a different data source, such as fitness science knowledge, weather API, session history. The LLM can then synthesise them into a single coherent message rather than presenting them as separate alerts.

5.3.4 Adaptive Plan Modification and Negotiation

Static training plans are a known retention problem: users miss sessions, life gets in the way, and the plan quickly falls out of sync with reality. LLMs enable a conversational re-planning flow where the user can explain a disruption like *"I had to skip this week's long ride because of work"*, and the model can reason about how to restructure the remaining plan, what to prioritise, and what to safely drop, rather than simply shifting everything forward by one week (the naive approach that most current apps take). This negotiated adaptation is a significant differentiator and maps directly to the Self-Determination Theory autonomy dimension discussed in Section 6.

5.4 Technical Implementation Considerations

5.4.1 Context Assembly and Memory Architecture

Because LLMs do not retain persistent user-specific memory across sessions and cannot access current athlete data from their weights alone, a dedicated context and memory layer is required. In a personalized running-coach system, this layer should combine structured data retrieval, long-term memory retrieval, and short-term session-state management rather than relying on semantic retrieval alone.

A typical implementation pattern is as follows:

1. The user submits a query, such as “Should I do my intervals today?”
2. The system retrieves relevant structured context from user databases and external tools, including variables such as current training load balance, recent sessions, today’s planned workout, HRV trends, and weather conditions.
3. Where relevant, the system also retrieves longer-term personalized memory, such as prior injury notes, coach observations, and summaries of previous training blocks.
4. A session-state layer, such as Durable Objects, maintains active per-athlete working memory, including recent conversation history, workflow progress, temporary coaching state, and other short-term decision variables.

This assembled context is then injected into the prompt so the model can generate advice grounded in the athlete’s actual state rather than generic population-level guidance. In practice, response quality depends heavily on the freshness, relevance, and structure of the retrieved context, making data design and memory architecture as important as model selection.

5.4.2 Model Selection

Approach	Example	Latency	Cost	Data privacy
API-hosted frontier model	GPT-4o, Sonnet	Low-moderate	Per-token usage	Data leaves your infra
Open-weight models, cloud hosted	Llama 3.1 70B on own infra	Moderate	Compute cost only	Data stays in your infra
On-device small model	Gemma 2B	Very low	Zero marginal cost	Fully private
Hybrid	On-device + API	Best of both	Mixed	Configurable

5.4.3 Prompt Engineering for Fitness Coaching

LLMs used as fitness coaches require careful system prompt design to produce consistently appropriate outputs. Key considerations include:

- The model should be instructed to cite the user's actual data when making recommendations, not generic advice, to build trust and explain reasoning
- A clear disclaimer layer is needed distinguishing general fitness guidance from medical advice — this is both a product safety concern and a regulatory requirement in many jurisdictions
- The model should be explicitly instructed to escalate to human review or recommend professional consultation when inputs suggest potential health risk (e.g. abnormally high resting HR, described chest pain, extreme weight loss goals)
- Responses should be calibrated to user expertise level — a beginner asking about heart rate zones needs a different explanation than a cyclist who already understands power zones and FTP

5.4.4 Limitations and Risks

LLMs in fitness coaching have several important limitations. They can hallucinate, producing confident but incorrect training or exercise science advice, which creates real safety risks unless outputs are grounded in reliable sources and numerical recommendations are generated from validated models rather than by the LLM itself.

There is also regulatory risk. AI-generated health and fitness guidance sits within an evolving legal framework, especially around health-related recommendations and the handling of personal data, which raises requirements for transparency, oversight, and data protection.

A further concern is user over-reliance. Highly automated coaching may reduce autonomy and intrinsic motivation over time, potentially making users passive rather than helping them build lasting self-management skills.

Finally, consistency across sessions is a challenge. Without explicit memory or retrieval mechanisms, an LLM will not remember prior conversations, which can lead to an unconvincing and frustrating coaching experience.

5.4.5 Recent Developments (2023-2025)

- Whoop Coach (2023):
  - The first major commercial deployment of an LLM coaching interface in a wearable fitness product. Users can ask natural language questions about their recovery score, sleep data, and strain. Initial reception was positive for explanation quality but criticised for shallow personalisation — the model did not learn from individual user patterns over time.
- Apple Intelligence integration with Fitness (2024):
  - Apple's on-device model framework enables privacy-preserving summarisation and coaching suggestions within Apple Fitness+, running entirely on-device via the Neural Engine on A17+ chips.
- Google Gemini + Fitbit (2024):
  - Google began integrating Gemini-based conversational features into Fitbit Premium, allowing users to query their health trends in natural language. Early reviews noted accuracy issues with complex multi-week trend analysis.
- Research:
  - LLM-generated training plans vs. coach-generated plans (Chen et al., 2024): A small-scale study (n=47) found that GPT-4-generated 12-week marathon training plans were rated comparably to coach-generated plans by recreational runners on perceived appropriateness, but significantly lower on perceived personalisation. This suggests that structured user data integration remains the critical gap.

5.4.6 Summary

LLMs are best used as a communication and reasoning layer on top, not a replacement for the quantitative personalisation models that underpin serious fitness analytics.

Their primary value in a running and cycling application is in translating structured data into meaningful, contextual, natural language guidance, handling the ambiguity and nuance of real user goals and situations, and enabling a coaching interaction that feels responsive rather than mechanical.

The combination of a well-structured RAG pipeline, a hybrid on-device/cloud model deployment, and careful prompt engineering to prevent hallucination and maintain appropriate scope represents the current state of the art and a genuine product differentiator against incumbent applications that have not yet integrated this capability.

6. Social Features, Gamification & Competition Mechanics

6.1 Social Mechanics in Fitness Apps

Social features in fitness apps serve two functions: they provide motivation through accountability and comparison, and they generate organic user acquisition through sharing behaviour.

The most effective social mechanic in running and cycling apps is Strava's segment system, which uses geofenced leaderboards applied to specific sections of public roads or trails. Any user who rides or runs through a segment automatically competes for the fastest recorded time, creating organic competitive engagement without requiring explicit challenge setup.

Feature	Mechanism	Gamification Element	Platform Example
Segments / KOM	Geofenced best-time leaderboard on a route section	Competition, achievement	Strava
Kudos / Likes	One-tap social affirmation on completed activities	Social influence	Strava
Challenges	Monthly distance or elevation targets shared by all users	Achievement, social proof	Strava, Nike Run Club
Virtual Races	Timed events run independently but ranked together	Competition, community	Zwift, Strava
Clubs & Group Feeds	Activity feed shared within an opt-in group	Belonging, accountability	Strava, Garmin Connect
Badges & Trophies	Earned icons for milestones (first 10K, 100-day streak)	Achievement, progress	Nike Run Club, Polar
Share Cards	Auto-generated activity summary graphic for social media	Social proof, virality	Strava, Garmin, Nike
Leaderboards	Ranked comparison of users within a group or challenge	Competition, status	Zwift, Strava

Table 6.1 — Social and gamification mechanics in fitness apps

6.2 Theoretical Framework: Self-Determination Theory

Self-Determination Theory (Deci & Ryan, 1985) provides the foundational psychological framework for understanding why gamification works in fitness contexts. The theory proposes three innate psychological needs that, when satisfied by an application, drive intrinsic motivation:

- Autonomy (The user's sense of agency over their training):
  - Satisfied by:
    - Customisable plans
    - Opt-in challenges
    - Flexible scheduling.
- Competence (The sense of growing ability):
  - Satisfied by:
    - Progress charts
    - Improving segment times
    - Adaptive plan difficulty.
- Relatedness (Connection to others):
  - Satisfied by:
    - Club feeds
    - Kudos
    - Shared challenges
    - Competition mechanics.

Apps that satisfy all three needs (Strava being the clearest example) achieve significantly higher long-term retention than those addressing only one dimension (e.g. apps that offer data tracking but no community features).

6.3 Social Media Integration and Privacy Considerations

Many major fitness apps integrate with Facebook, Instagram, and X (formerly Twitter) for activity sharing. The common integration pattern is that, post-session, a summary card (time, distance, map, pace) is generated and the user can share it directly. However, privacy concerns are significant:

- GPS route data embedded in share cards can inadvertently reveal a user's home location (the start/end of most runs).
- Strava's segment leaderboards created documented privacy incidents where users' precise home addresses were inferred from public data.
- Best practice — which should be built into the proposed system from day one — is to offer a 'privacy zone' feature that obscures the start and end of routes within a user-defined radius.

7. Business Model & SaaS Design

7.1 Business Model Comparison

Model	Example	Revenue Mechanism	Pros	Cons
Device-bundled free	Garmin Connect	Hardware margin funds app development	High retention; loyal hardware users	Locked to device ecosystem; zero software revenue
Pure freemium	Strava (free tier)	Large free user base, convert <10% to paid	Large addressable market; viral growth	Feature gatekeeping creates frustration; high churn on paid
Subscription-first	Wahoo SYSTM, Zwift	Monthly/annual recurring revenue	Predictable revenue; strong incentive to deliver value	High barrier to trial; price sensitivity
Freemium + social lock-in	Proposed model	Free tracking + paid coaching/analytics; social features free to drive growth	Best of both: viral growth + upsell path	Must clearly justify premium value proposition
B2B / Wellness platform	Wellhub (Gympass)	Corporate health programme contracts	High contract value; stable revenue	Long sales cycles; not relevant for consumer launch

Table 7.1 — Business model comparison for fitness SaaS applications

7.2 Recommended Model: Freemium with Social-Led Growth

For the proposed system, a freemium model with strong free social features is recommended. This mirrors Strava's approach but addresses its key weakness — insufficient value in the free tier for users who do not care about segment leaderboards.

Feature	Free Tier	Premium Tier ($9.99/mo)
Activity tracking (GPS, HR)	✓ Unlimited	✓ Unlimited
Basic analytics (pace, distance, HR zones)	✓ Last 30 days	✓ Full history
Social feed, kudos, clubs	✓ Full access	✓ Full access
Challenges & competitions	✓ Full access	✓ Full access
AI coaching & training plans	Basic (3 workouts/week)	Full adaptive plans

Feature	Free Tier	Premium Tier ($9.99/mo)
ATL/CTL/TSB training load dashboard	✗	✓
Advanced analytics (VO2max trend, HRV)	✗	✓
Predictive race time estimates	✗	✓
Route creation & planning	✗	✓
Priority alert system (overtraining)	✗	✓

Table 7.2 — Proposed feature tier allocation

8. Implementation, Tech Stack & Cross-Platform Strategy

8.1 Cross-Platform Framework Comparison

Framework	Language	Performance	BLE / Sensor Access	Background GPS	Community	Recommendation
React Native	JavaScript / TypeScript	Good	Via react-native-ble-plx	Via background geolocation library	Very large	Strong choice for JS teams
Flutter	Dart	Near-native	Via flutter_blue_plus	Via background_locator	Large, fast-growing	Recommended — best perf/DX balance
Kotlin Multiplatform	Kotlin	Native	Direct platform APIs	Direct platform APIs	Emerging	Best for Kotlin-experienced teams
Ionic / Capacitor	JavaScript / TypeScript	Moderate	Via community plugins	Limited reliability	Moderate	Not recommended for fitness apps
Native (Swift + Kotlin)	Swift / Kotlin	Native	CoreBluetooth / BluetoothGatt	Full platform support	Platform-native	Best performance, 2x codebase cost

Table 8.1 — Cross-platform framework comparison for fitness app development

Flutter is the recommended framework for the proposed system. It provides near-native performance for GPS and BLE functionalities, enables cross-platform development through a single codebase for iOS and Android, and is supported by a strong community with a wide range of plugins suitable for fitness applications.

8.2 Alert System Design

The proposed alert system operates across four distinct trigger types, each targeting a different user behaviour pattern:

Alert Type	Trigger Condition	Delivery Method	Example Message
Scheduled reminder	No activity logged by user's typical workout time + 2 hours	Push notification	'It's your usual run time — ready to go?'
Goal-proximity alert	User within 500m / 10 min of daily or weekly distance goal	Push notification	'Only 0.8km left to hit your weekly goal!'
Overexertion warning	HR in Zone 5 for >3 continuous minutes during a session	In-app vibration + banner	'Your HR has been very high for 3+ minutes. Consider slowing down.'

Alert Type	Trigger Condition	Delivery Method	Example Message
Recovery recommendation	TSB (Training Stress Balance) drops below −30	Push notification (next morning)	'Your training load is high. Consider an easy day or rest today.'
Inactivity nudge	No activity logged in past 3 days (if plan active)	Push notification	'You've not trained in 3 days — your plan has an easy run today.'
Weather-adaptive suggestion	Rain or extreme heat forecast during scheduled session time	Push notification (2h before)	'Heavy rain forecast at 6pm. Shift your run to this morning?'

Table 8.2 — Alert system types, triggers, and example messages

8.3 Common Software Tools & Libraries

Category	Tool / Library	Purpose
Mobile framework	Flutter + Dart	Cross-platform iOS/Android development
BLE connectivity	flutter_blue_plus	BLE GATT device connection and data streaming
GPS tracking	background_locator, geolocator	Foreground and background GPS session recording
Maps	Mapbox SDK or Google Maps Flutter	Route visualisation and segment display
Backend API	Node.js + Express or Python FastAPI	REST API for data ingestion, user management
Analytics engine	Python (NumPy, Pandas, scikit-learn)	Training load calculation, ML model serving
Database	PostgreSQL + TimescaleDB extension	Time-series activity data at scale
Cloud infrastructure	AWS (ECS, RDS, Lambda) or GCP	Scalable SaaS backend
Notifications	Firebase Cloud Messaging (FCM)	Cross-platform push notification delivery
Social sharing	Facebook SDK, Instagram Graph API	Share card generation and posting
Auth	Auth0 or Firebase Auth	OAuth2 social login (Google, Apple, Facebook)

Table 8.3 — Recommended software tools and libraries

9. Case Studies & System Modelling

9.1 Case Study: Strava Segment Engine

Strava's segment engine is the most cited example of organic competitive gamification in fitness software. Segments are geofenced sections of roads or trails, defined by GPS start and end coordinates. When any user travels through a segment — without needing to know it exists — their time is automatically recorded and added to the segment's public leaderboard.

Impact metrics: Strava reported over 100 million segments created globally by 2023, with segment leaderboards cited by users as one of the primary drivers of app re-engagement (Strava Year in Sport Report, 2023). The segment model creates emergent competition without requiring users to actively initiate a challenge — a key design principle for reducing friction in social engagement.

Key design lesson for the proposed system: competition features should be ambient and automatic where possible. Requiring users to manually find opponents and send challenge invites adds friction and reduces participation. Automatic leaderboards seeded by normal activity behaviour achieve higher engagement.

9.2 Case Study: Garmin Connect Training Status & Body Battery

Garmin's Training Status feature combines VO2max estimation, training load (7-day and 4-week), and HRV-derived recovery data into a single daily readiness signal. Body Battery extends this by incorporating sleep quality and overnight HRV into a 0-100 energy score. Both are calculated on-device by Firstbeat Analytics algorithms.

Key design lesson: integrating multiple data streams (HR, HRV, sleep, training load) into a single interpretable readiness score is significantly more actionable for users than presenting each metric separately. The proposed system should adopt a similar single-number readiness score built from available data sources.

9.3 Case Study: Zwift — Gamification at Scale

Zwift transformed indoor cycling training by creating a persistent virtual world (Watopia) where users ride avatars through 3D environments. Real-world power output from smart trainers controls avatar speed in real time. The platform hosts structured workouts, group rides, and full racing leagues with divisions and season rankings.

By March 2024, Zwift reported over 4 million registered users across 190 countries. Key success factors: real-time interaction (seeing other riders' avatars creates genuine pack dynamics), structured progression (racing categories prevent sandbagging), and community events (Tour de Zwift, Zwift Academy) that create periodic re-engagement spikes.

Key design lesson for running/cycling: real-time or near-real-time competition — even when participants are physically separate — is a powerful retention mechanic. The proposed system should support live competition modes (virtual races) not just post-hoc leaderboards.

9.4 User Journey Map

The following table maps the key stages of a user's journey from initial discovery to long-term engagement, identifying the touchpoints, user goals, and system responsibilities at each stage.

Stage	User Goal	Key Touchpoints	System Responsibilities	Success Metric
Discovery	Find a solution for tracking training	Marketing website, App Store listing, social media share cards from existing users	Clear value proposition, compelling screenshots, social proof	App Store conversion rate
Onboarding	Set up profile and connect devices	Welcome flow, goal-setting screen, BLE pairing wizard	Frictionless BLE pairing, clear goal options, instant value delivery	Onboarding completion rate
First session	Record first run or ride	GPS lock screen, live stats display, session summary	Reliable GPS start, real-time metrics, celebratory summary card	Session completion rate
Social discovery	Find friends, join clubs	Friend finder, suggested clubs, first kudos received	Suggest relevant clubs, prompt friend connections via contacts/social auth	Social connection rate within 7 days
Plan engagement	Start a structured training plan	Plan selection screen, first scheduled workout notification	Appropriate plan recommendation based on profile, reminder system	Plan start rate
Competition	Win a segment or challenge	Leaderboard view, competition invite, achievement notification	Accurate segment timing, fair ranking, timely notifications	Competition participation rate
Premium conversion	Access advanced analytics	Paywall in analytics section, free trial prompt	Clear demonstration of premium value, easy payment flow	Conversion rate (target: 8–12%)
Long-term retention	Maintain fitness progress	Weekly summary, plan updates, seasonal challenges	Adaptive plan adjustments, consistent engagement triggers	D90 retention rate

Table 9.1 — User journey map from discovery to long-term retention

10. Proposed System, Research Gaps & Conclusion

10.1 Target Users

The proposed system targets recreational and amateur endurance athletes, specifically:

- Recreational runners aged 18–45 who complete 2–5 runs per week, have a goal (e.g. finish a 10K or half marathon), but lack the time or knowledge to structure their own training.
- Amateur cyclists who ride for fitness or club sportive events, own a modern Bluetooth-enabled device, and want actionable post-ride analysis beyond basic GPS mapping.
- Beginners who have recently started running or cycling and need guided onboarding, structured plans, and motivational social features to build the habit.

Elite and professional athletes are explicitly out of scope for the initial product — their needs (lab-grade data, coach collaboration tools, periodisation software) are better served by specialist platforms such as TrainingPeaks and Today's Plan.

10.2 Research Gap Analysis

The following table synthesises the review findings into a structured gap analysis, directly mapping what existing applications offer against what is missing and what the proposed system should address.

Research Area	What Existing Apps Do	What Is Missing	Proposed System Response
Profile-based recommendations	Garmin and Wahoo offer strong profiling; Strava and Nike offer basic goal plans	Free-tier AI coaching is absent across all major platforms	AI coaching included in free tier (limited sessions); full adaptive plans in premium
Data collection	BLE and ANT+ well supported; GATT profiles standardised	Multi-protocol aggregation in a single session is cumbersome; no unified sensor dashboard	Unified BLE + ANT+ session with automatic sensor discovery and role assignment
Social & competition	Strava's segment engine is industry gold standard	Real-time competition (live virtual races) absent from running apps	Live competition mode with real-time pace comparison against registered opponents
Alert system	Basic scheduled reminders and workout completion notifications common	Predictive overtraining alerts and recovery-based nudges rare in free tier	Full ATL/CTL/TSB-based alert system in premium; basic HR zone alerts in free

Research Area	What Existing Apps Do	What Is Missing	Proposed System Response
Cross-platform UX	Most apps support iOS and Android but UI quality varies	Consistent UX across phone sizes and OS versions is inconsistently achieved	Flutter single codebase with responsive layout system and design token system
Business model	Freemium dominant; premium value often insufficiently justified	Premium tiers often gate social features, reducing viral growth	Social features fully free; premium gates analytics and AI coaching only
Predictive analytics	Strava Premium offers fitness/freshness chart; Garmin offers training status	Race time prediction and adaptive plan generation in free tier nonexistent	ML-powered race time estimator and adaptive weekly targets in premium

Table 10.1 — Research gap analysis: existing apps vs proposed system

10.3 Intended Functionalities of the Proposed System

Based on the literature review findings, the proposed system will implement the following prioritised feature set:

Priority	Feature	Informed By
P1 — Core	GPS and BLE session recording (running and cycling)	Universal requirement; GATT standards review (Section 3)
P1 — Core	Real-time metrics display (pace, HR, cadence, power)	User expectation set by all existing apps
P1 — Core	Post-session analysis with zone breakdown and map	Competitor standard (Strava, Garmin)
P1 — Core	User profile creation with goal-setting and fitness baseline	Section 4 profiling model
P2 — Growth	Social feed, kudos, clubs, and challenge system	Strava case study (Section 8.1); SDT framework (Section 5.2)
P2 — Growth	Automatic segment leaderboards on popular local routes	Strava case study — ambient competition (Section 8.1)
P2 — Growth	Activity share cards for Instagram, Facebook, and X	Competitor standard; virality driver
P3 — Premium	ATL/CTL/TSB training load dashboard	Section 4.2; Garmin/TrainingPeaks model
P3 — Premium	Adaptive training plan generation and modification	Section 4.4 ML approaches

Priority	Feature	Informed By
P3 — Premium	VO2max tracking and race time prediction	Section 4.3; Firstbeat algorithm model
P3 — Premium	Overtraining and recovery alert system	Section 7.2 alert design; Garmin Body Battery case study

Table 10.2 — Prioritised feature plan with literature justification

10.4 How to Implement: Algorithm and Approach Summary

The following approach is recommended based on the literature review:

- Phase 1 — Data collection: Implement a GATT-compliant BLE central that aggregates Heart Rate Service, Cycling Power Service, and Running Speed and Cadence Service streams simultaneously, with GPS from device location APIs. Store raw streams in FIT-compatible JSON format.
- Phase 2 — Analytics: Implement the ATL/CTL/TSB model server-side using exponential weighted moving averages (τ=7 days for ATL, τ=42 days for CTL). Calculate TSS per session using HR-based Intensity Factor for users without power meters.
- Phase 3 — Personalisation: Train a collaborative filtering model on anonymised session data to generate workout recommendations. Fine-tune with a lightweight LSTM to predict weekly performance trajectory.
- Phase 4 — Social: Implement geofenced segments using Haversine-distance matching of GPS routes against a PostGIS database of registered segment coordinates. Compute segment times server-side to prevent client-side manipulation.

10.5 Recent Developments in the Topic Area

The following developments in the 2022–2025 period are particularly relevant to the proposed system:

- Large language models as fitness coaches: GPT-4 class models have been prototyped as conversational coaching interfaces that answer training questions in natural language, synthesising a user's history with general exercise science knowledge. Integration as a conversational layer over the analytics backend is a strong product differentiator.
- On-device ML inference: Apple's Core ML and Google's ML Kit now enable lightweight model inference on-device, reducing latency for real-time coaching suggestions and enabling offline operation.
- HRV-based readiness scoring: Consumer HRV measurement via wrist PPG has matured sufficiently (within ±10ms of chest strap in controlled conditions) to power recovery recommendations without additional hardware.
- EU AI Act (2024) implications: AI-generated health coaching recommendations may create additional disclosure, documentation, oversight, and data-governance requirements in EU markets.

10.6 Conclusion

This literature review has systematically addressed all research questions set out in the project brief. The review confirms that:

1. A significant and growing market exists for running and cycling fitness applications, with no single mainstream platform clearly combining AI personalisation, social competition, and cross-platform accessibility.
2. Data collection is technically mature — BLE GATT profiles and ANT+ provide a standardised, interoperable sensor layer that applications can build upon with established libraries.
3. The ATL/CTL/TSB training load model, combined with VO2max estimation and emerging ML approaches, provides a solid algorithmic foundation for personalised coaching recommendations.
4. Strava's segment engine remains the gold standard for ambient social competition and should inform the design of the proposed system's competition mechanics.
5. A freemium business model with social features in the free tier and analytics/coaching in premium is best suited to driving viral growth while generating sustainable SaaS revenue.
6. Flutter is the recommended implementation framework, offering near-native performance, a single codebase for iOS and Android, and strong community support for fitness-specific sensor integration.

The proposed system has a clearly differentiated position: AI-powered coaching accessible in the free tier, full social competition features for all users, and a premium analytics layer built on established endurance sport science. This combination is not clearly offered in this form by a single mainstream competitor.

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